PASSWORD VAULT AND CREDENTIAL STORAGE SYSTEM

Java Assignment

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INTRODUCTION

A password vault or credential storage system is a secure digital tool designed to help individuals and organizations manage their authentication data—such as usernames, passwords, API keys, and certificates—in a centralized, encrypted environment.

OBJECTIVE

- It is developed using core java
- It helps users to view, delete login credentials like website name,username and password
- It is a console based java application

MAIN PARTS

It has two main parts

- **Credential class :**

  It stores data like site,user name and password

- **Main Class**

  It uses an arraylist to store all credentials

Concept

- Class
- Object
- Arraylist
- Loop(do while)
- Iterator

Uses

- Secure storage of usernames, passwords, and sensitive credentials in encrypted form.
- Centralized management of multiple accounts across devices and platforms.
- Automatic password generation to ensure strong, unique credentials.
- Cross-device synchronization for seamless access on mobile, desktop, and web.

- **Credrntial:**
  - It store user data like website name,user name,password
  - Each credential is stored as an object
- **Menu deiven program:**
  - It displays options to user
  - And it performs according to user choice
- **ArrayList:**

  It allows adding and removing data easily without any fixed size

FUNCTIONS

- **Add Credential**

Takes input from the user and stores it in arraylist .If you want to add or login another account you can add account

- **View Credential**

It displays all stored cedentials .If it is empty ,it shows message

- **Delete Credential**

It searches by website name and removed matching credential .If you want to delete youn can delete

ADVANTAGE

- Secure storage with strong encryption
- Easy access using one master password / MFA
- Reduces password reuse and weak passwords
- Centralized management for users & enterprises

DISADVANTAGE

- Single point of failure (if master password is lost)
- Risk if vault itself is hacked
- Requires user trust in the system/provider
- May need regular updates and maintenance

Future scop

- Biometric login – fingerprint, face ID instead of master password.
- Cloud vaults – access anywhere, anytime.
- Zero-knowledge design – even provider can’t see your data.
- IoT integration – manage smart device logins.